Telecom Portfolio Risk Report

Generated 12 August 2026 · portfolio state, not a time series

Unmarked	OBSERVED - measured from the source data.
†	PROJECTED - derived from the stated economic assumptions, not measured.
‡	SIMULATED - from generated history. Shows portfolio structure only; churn in it is flat by construction and cannot support a forecast.

Portfolio at a glance

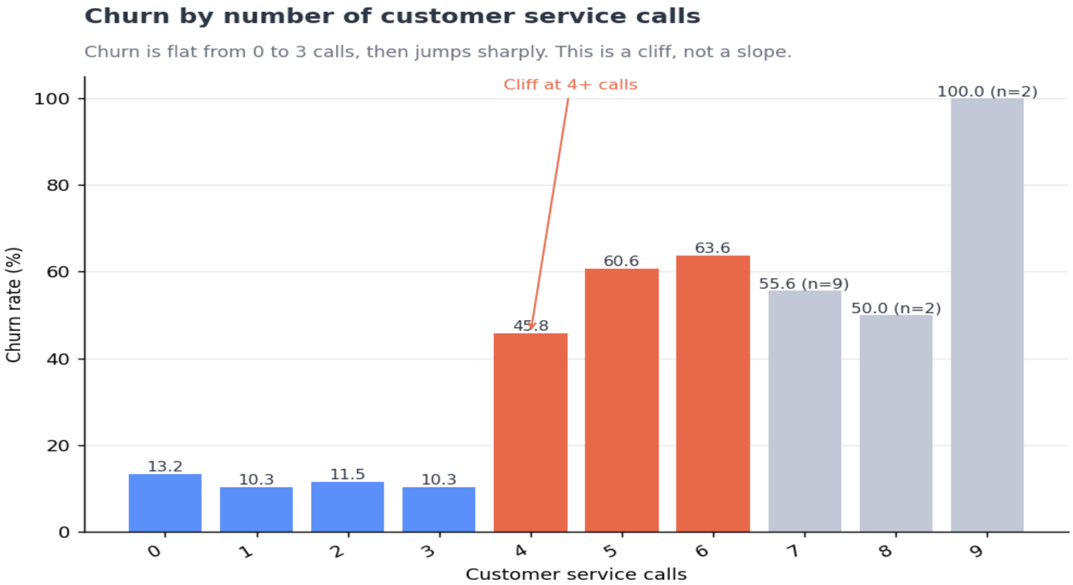
3,333	**14.49%**	**483**
Customers	Churn rate	Churned
$198,146	**$59.45**	**$31,567**
Revenue	ARPU	Revenue at risk

The most decision-relevant fact is that the portfolio has a churn rate of 14.49%, resulting in revenue at risk of 31566.93. The confirmed churn drivers are 4+ customer service calls, international plan subscription, and day charge greater than or equal to 45. The customer group that needs attention first is the 267 high service call customers, as they are at a higher risk of churn due to exceeding the service call cliff of 4 or more calls. Additionally, the 323 international plan subscribers and 210 heavy day usage customers are also at risk. The total revenue of 198146.03 is at stake, with the average revenue per user being 59.45. To mitigate churn, focusing on these high-risk groups is essential.

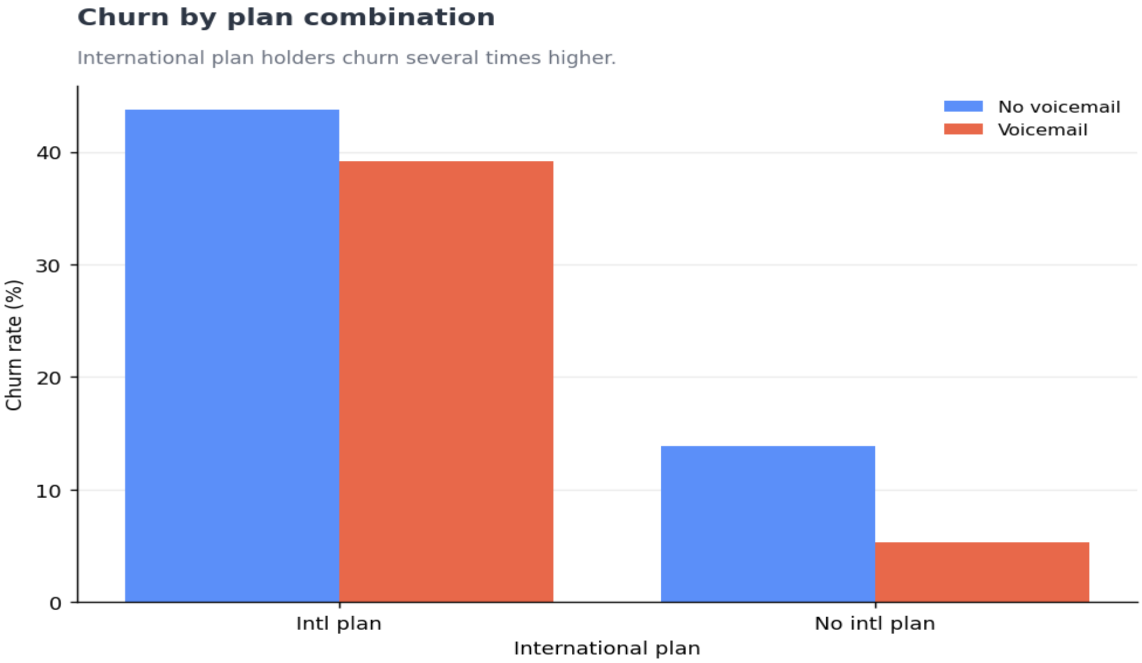
Confirmed churn drivers

Narration unavailable: Groq rate limit after 5 attempts. Lower concurrency or raise LLM_MIN_INTERVAL. *{"error":{"message":"Rate limit reached for model `llama-3.3-70b-versatile` in organization `org_01kyrzzy9df89bsznh3c4agqa5` service tier `on_demand` on tokens per day (TPD): Limit 100000, Used 99428, Requested 1936. Please try again in 19m38.495999999s. Need more tokens? Upgrade to Dev Tier today at https://console.groq.com/settings/billing","type":"tokens","code":"rate_limit_exceeded"}}*

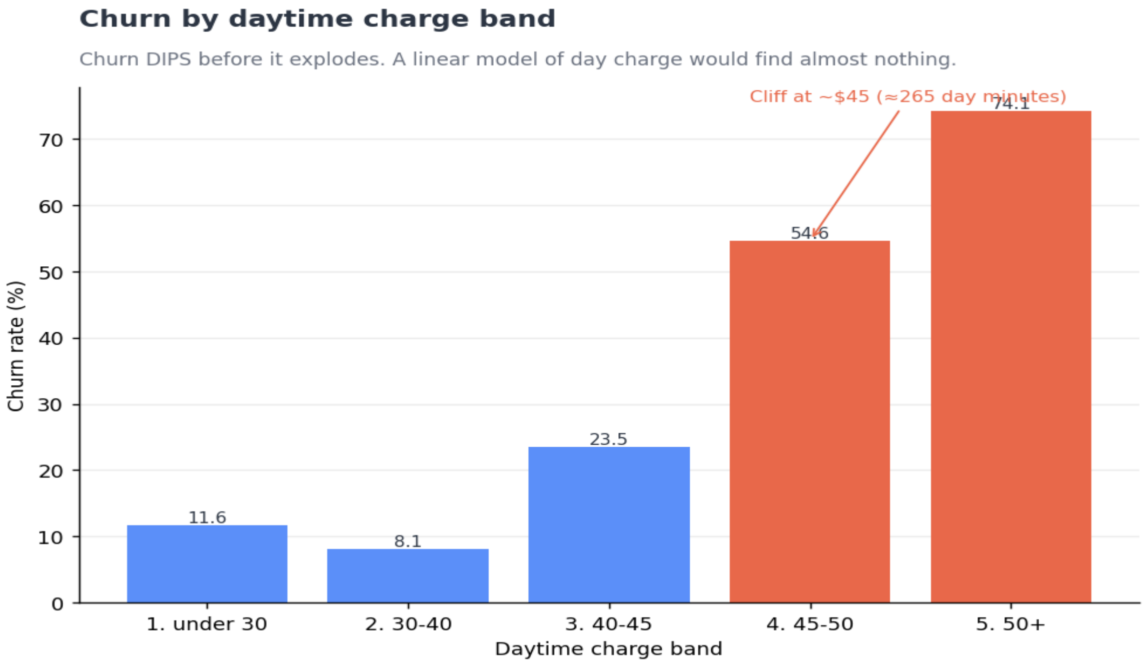
Driver	Threshold	Below	Above	Customers
Customer service calls	4 or more	11.25%	51.69%	267
International plan	subscribed	11.5%	42.41%	323
Daytime charge	$45 or more	11.43%	60.0%	210



Bucket as '0-3' vs '4+'. Treating call count as a continuous predictor understates the effect badly. Greyed bars have fewer than 20 customers and are too small to read as evidence.



Voicemail plan has little independent effect; the international plan is the driver.



Found while building the Phase 6 rules engine. The earlier two-driver segmentation reported a 'baseline' of 8.2% that concealed this group.

Risk segments and recommended actions

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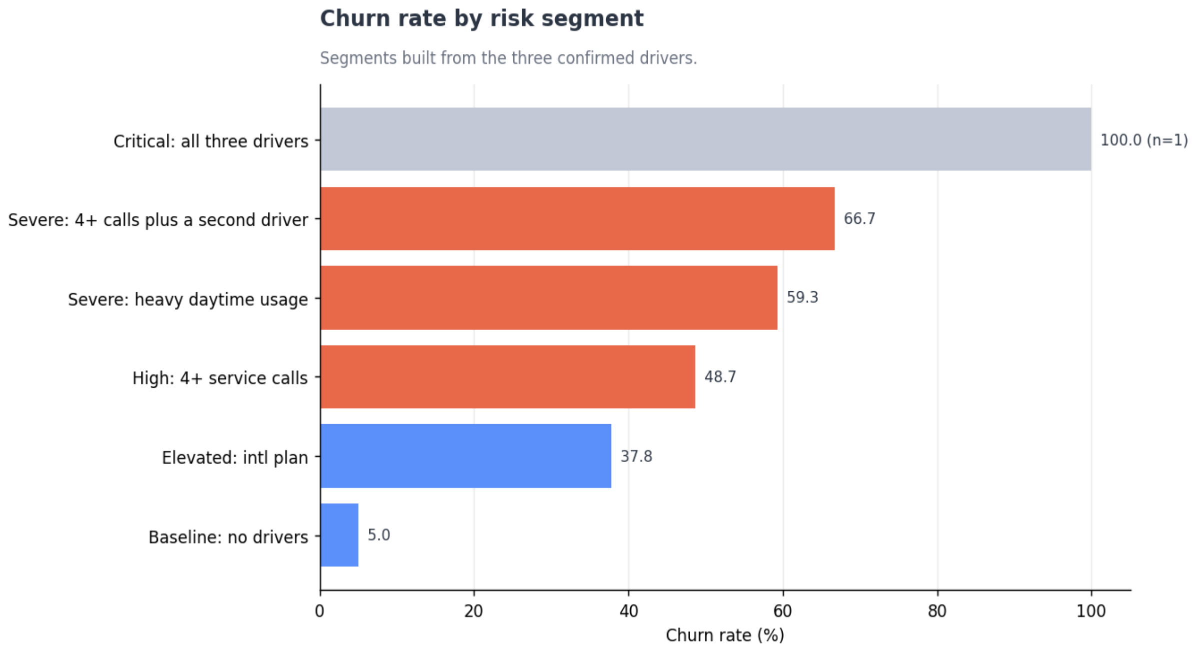
Segment	Customers	Churn	Revenue at risk	Net value†
Critical — all three risk drivers	1	100.0%	$82	$242†
Severe — 4+ service calls with a second driver	42	66.67%	$1,834	$5,752†
Severe — heavy daytime usage	194	59.28%	$9,226	$29,017†
High — 4+ service calls only	224	48.66%	$5,549	$20,044†
Elevated — international plan only	267	37.83%	$6,139	$20,823†
Baseline — no risk drivers	2,605	4.95%	$8,738	-

Recommended actions

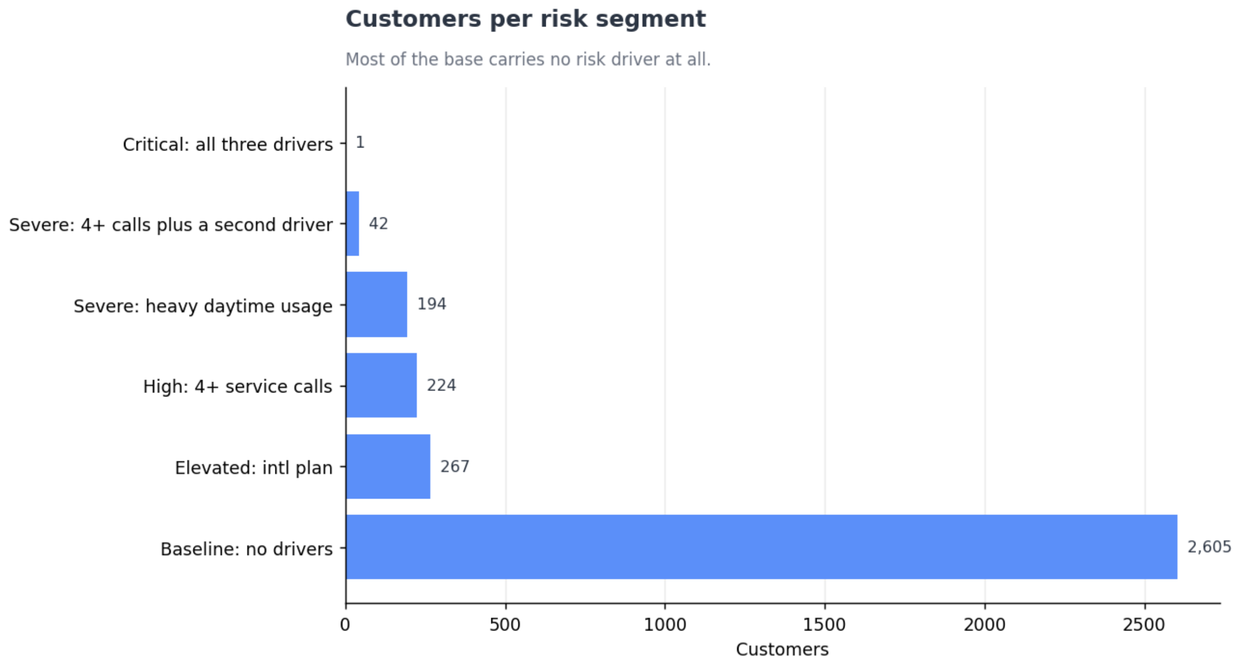
Segment	Action	Offer	Customers
Critical — all three risk drivers	Immediate senior retention call, same day	Premium support + tariff review + 20% discount, 3 periods	1

Segment	Action	Offer	Customers
Severe — 4+ service calls with a second driver	Senior retention specialist call within 48 hours	Premium support tier + 15% discount for 3 periods	42
Severe — heavy daytime usage	Tariff review call within 5 working days	Move to a daytime bundle or higher-allowance plan	194
High — 4+ service calls only	Proactive support review call within 5 working days	Service credit + dedicated support contact	224
Elevated — international plan only	Plan review — check the tariff fits actual usage	Tariff optimisation or downgrade without penalty	267

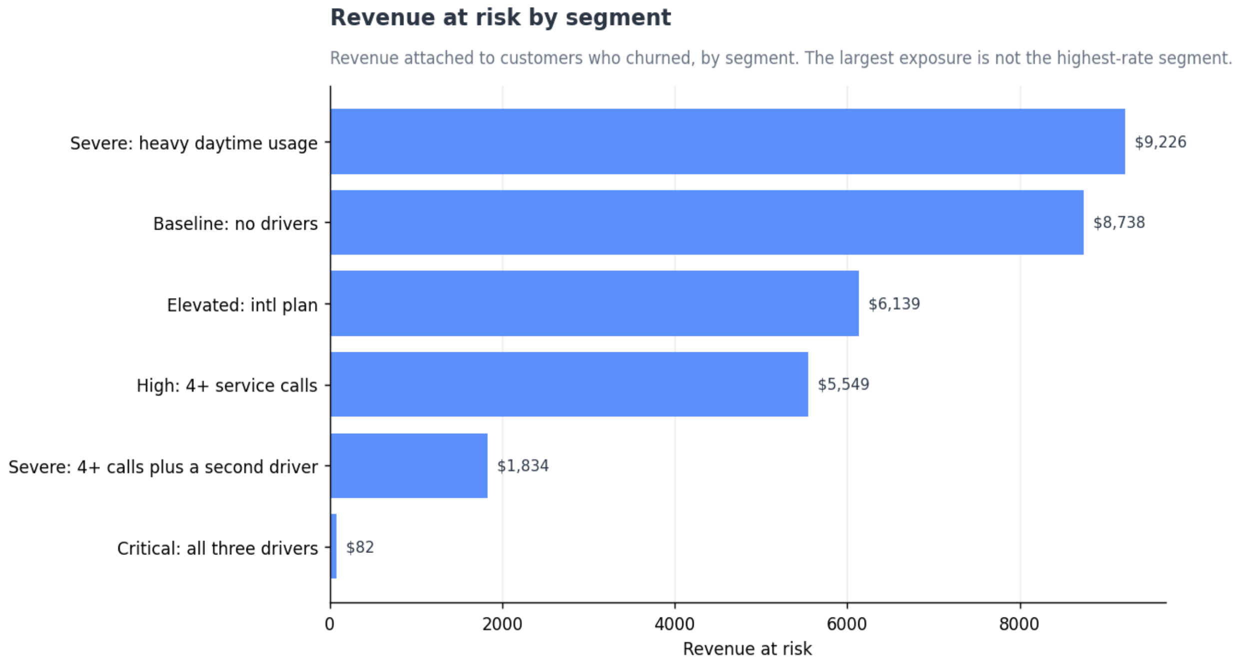
† **Assumptions behind projected values:** save rate 30% (band 15%-45%), contact cost $8.00, acquisition cost $45.00, horizon 12 periods. These are industry-typical placeholders, not measurements: the dataset contains no cost data.



Segments are mutually exclusive and exhaustive: counts sum to 3,333 and revenue to the portfolio total. Greyed bars have fewer than 20 customers and are too small to read as evidence.



Segment sizes are highly uneven; read rates alongside n.



Rate and exposure rank differently. A large low-rate segment can carry more absolute revenue risk than a small high-rate one.

Tenure: an expected driver that is not one

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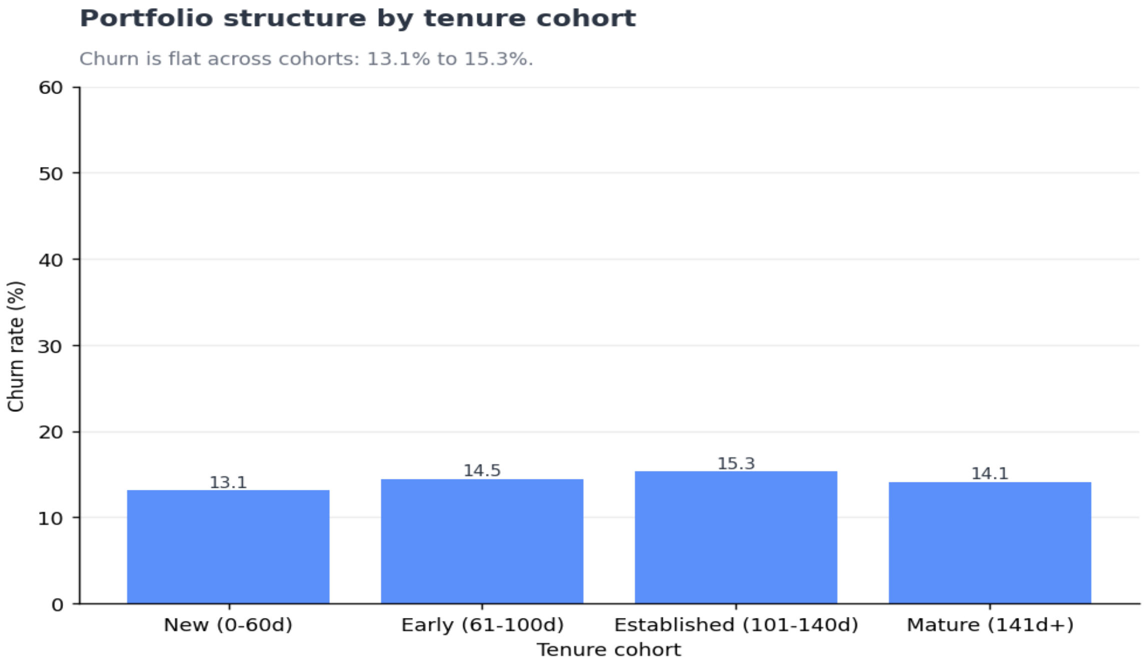
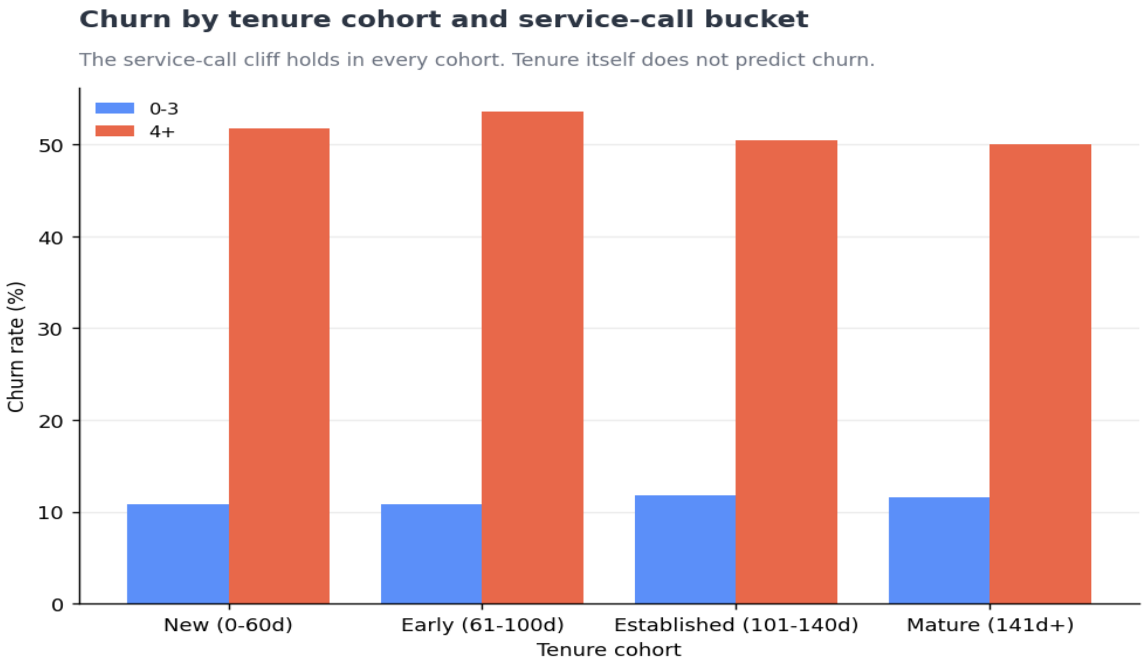
Cohort	Customers	Churn	ARPU	Avg service calls
New (0-60d)	511	13.11%	$59.70	1.52

Cohort	Customers	Churn	ARPU	Avg service calls
Early (61-100d)	1,153	14.48%	$59.13	1.57
Established (101-140d)	1,138	15.29%	$59.49	1.6
Mature (141d+)	531	14.12%	$59.83	1.5



Plotted on the same y-scale as the driver charts to show how little variation there is here by comparison.



Tenure is the ordered axis in this project because the dataset has no date column. It is useful for structure, not as a churn driver.

Portfolio structure over time‡

SIMULATED HISTORY. This section is derived from generated monthly snapshots, not observed history. It shows portfolio structure only. Churn in this panel is flat by construction and cannot support a trend claim or a forecast.

12	**569-654**	**$8,529-$9,954**
Months shown	Active base	Monthly revenue

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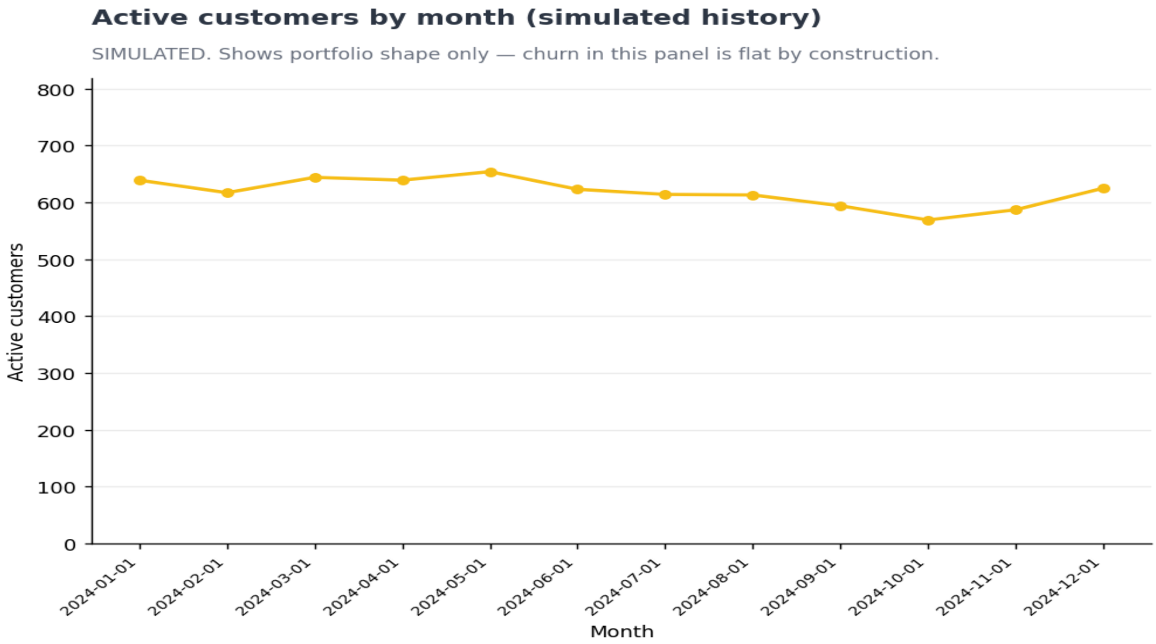
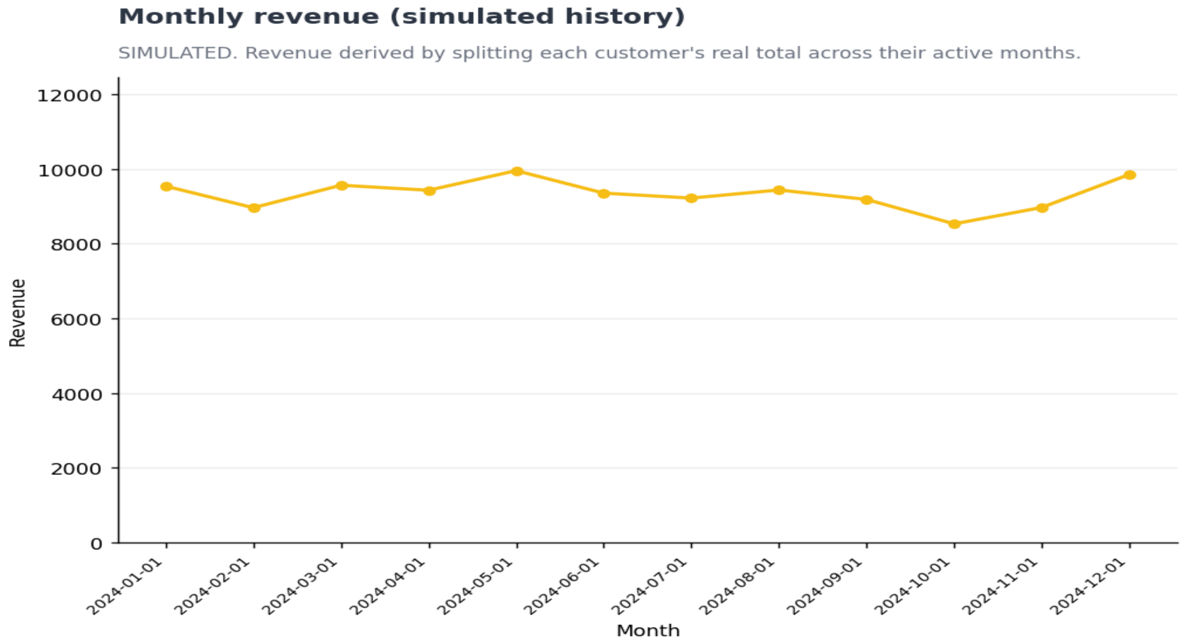
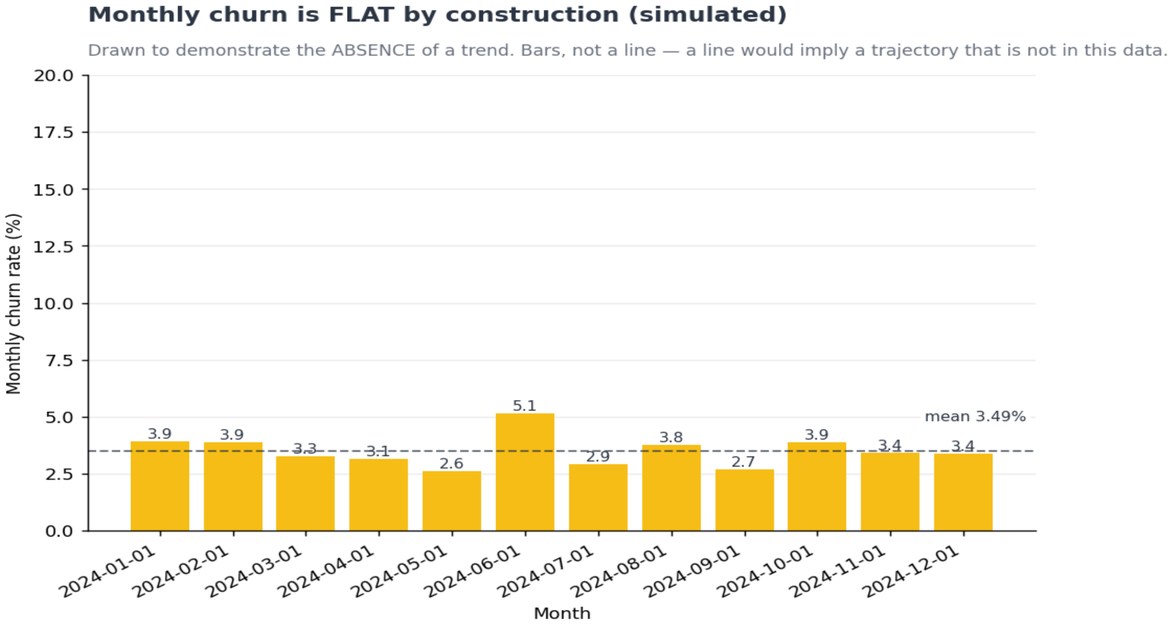


Generated from the snapshot, not observed. Included to show portfolio shape; it contains no information the snapshot did not already hold.



Each customer's real total charge was distributed across their active months. Monthly totals sum back to the real portfolio revenue.



Included to demonstrate the absence of a trend, not to show one. Churn timing in this panel was generated; a different random seed would produce a different arrangement of the same flat distribution. Real churn movement over time cannot be measured from a single snapshot.

What this report cannot tell you

• NO TIME DIMENSION. The source data is a single snapshot with no dates. Nothing here describes how any metric changed over time, and no figure in this report is a forecast.

• SIMULATED HISTORY SHOWS STRUCTURE, NOT TRENDS. The monthly panel in this report is generated from the snapshot, so it contains no information the snapshot did not already hold. Churn in it is deliberately flat: a trend there would reflect the generator's random seed, not customer behaviour. It is included to show portfolio shape and tenure lifecycle, both of which are legitimate uses of derived monthly data.

- TENURE IS NOT A DRIVER. Churn is flat across tenure cohorts (13.1%-15.3%). This was expected to matter and does not - a genuine finding, and a reason not to target by tenure.
- STATE-LEVEL FIGURES ARE NOISY. 51 states average about 65 customers each, so a handful of churners moves a state several percentage points. Geography is the weakest dimension here.
- PROJECTED VALUES REST ON ASSUMPTIONS. Save rate, contact cost and acquisition cost are industry-typical placeholders, not measurements: the dataset contains no cost data. A save rate of 30% is assumed, with a sensitivity band of 15%-45%.
- REVENUE IS PERIOD CHARGES. Not monthly recurring revenue and not annualised.
- CHURN RISK IS NOT PREDICTED HERE. Segments describe which customers resemble those who already churned. Scoring individual customers is a separate modelling problem.

Provenance

Every observed figure traces to the source data. Segment totals reconcile to the portfolio: customers match (3,333), revenue matches ($198,146.03).

Warning: narration in these sections failed validation and may contain unverified figures: drivers, segments, cohorts, structure.